$$20^2 = 16^2 + x^2$$

$$\Rightarrow 400 = 256 + x^2$$

$$\Rightarrow \quad x^2 = 400 - 256 = 144 = 12^2 \quad \Rightarrow \quad x = 12 \text{ feet}$$

Therefore, the base (foot) of the ladder is 12 feet away from the wall.

Activity

1. We can construct sets of Pythagorean triplets as follows.
   Let **m** and **n** be any two positive integers (m > n):
   (a, b, c) is a Pythagorean triplet if $a = m^2 - n^2, b = 2mn$ and $c = m^2 + n^2$ (Think, why?)
   Complete the table.

m	**n**	$a = m^2 - n^2$	$b = 2mn$	$c = m^2 + n^2$	Pythagorean triplet
2	1	___	___	___	_____
3	2	___	___	___	_____
4	1	15	8	17	(15, 8, 17)
7	2	45	28	53	(45, 28, 53)

2. Find all integer-sided right angled triangles with hypotenuse 85.

Example 5.14

Find LM, MN, LN and also the area of ΔLON.

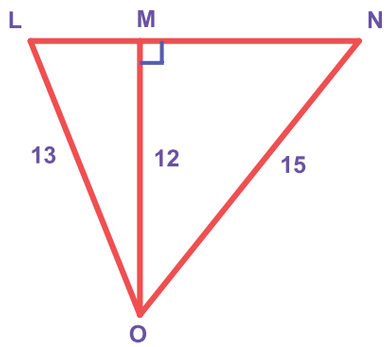


Fig. 5.27

Solution:

From ΔLMO, by Pythagoras theorem,

$$LM^2 = OL^2 - OM^2$$

$$\Rightarrow LM^2 = 13^2 - 12^2 = 169 - 144 = 25 = 5^2$$

$$\therefore \quad LM = 5 \text{ units}$$

From ΔNMO, by Pythagoras theorem,

$$MN^2 = ON^2 - OM^2$$

$$= 15^2 - 12^2 = 225 - 144 = 81 = 9^2$$

$$\therefore \quad MN = 9 \text{ units}$$

Hence, $LN = LM + MN = 5 + 9 = 14$ *units*

$$\text{Area of } \Delta LON = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times LN \times OM$$

$$= \frac{1}{2} \times 14 \times 12 = 84 \text{ square units.}$$

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